

Coding Theory and Entropy

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What is Information?

Intuition: Information = surprise / uncertainty

Example

Two messages:

- ▶ “The sun will rise tomorrow” → low information (expected)
- ▶ “A meteor will hit Earth tomorrow” → high information (surprising)

Properties of an information measure $S(p)$:

- 1 $S(p)$ decreases as p increases (less surprise)
- 2 $S(1) = 0$ (certain event gives no information)
- 3 For independent events: $S(p_1 p_2) = S(p_1) + S(p_2)$

Solution: $S(p) = \log_2 \left(\frac{1}{p} \right) = -\log_2 p$ (bits)

Shannon Entropy, Definition

Definition: For a random variable X with outcomes $x_1, \dots, x_n$ and probabilities $p_1, \dots, p_n$:

$$H(X) = \sum_{i=1}^n p_i \log_2 \left(\frac{1}{p_i} \right) = - \sum_{i=1}^n p_i \log_2 p_i$$

Interpretation:

- ▶ Average information (surprise) per symbol
- ▶ Minimum average number of bits needed to encode X
- ▶ Measures uncertainty or randomness

Units: bits (if base 2)

Entropy Examples

Fair coin:

$$p_H = p_T = \frac{1}{2}$$

$$H = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1 \text{ bit}$$

Biased coin:

$$p_H = 0.9, \quad p_T = 0.1$$

$$H \approx 0.469 \text{ bits}$$

Certain event:

$$p = 1$$

$$H = -1 \cdot \log_2 1 = 0 \text{ bits}$$

Uniform over 8 symbols:

$$p_i = \frac{1}{8}, \quad i = 1, \dots, 8$$

$$H = -\sum_{i=1}^8 \frac{1}{8} \log_2 \frac{1}{8} = \log_2 8 = 3 \text{ bits}$$

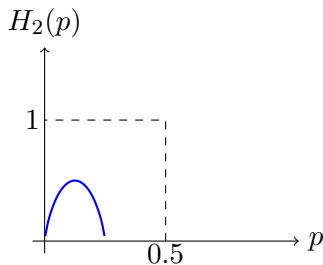
Binary Entropy Function

For binary $X \in \{0, 1\}$ with $\mathbb{P}(X = 1) = p$:

$$H_2(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$$

Properties:

- ▶ $H_2(0) = H_2(1) = 0$
- ▶ $H_2(\frac{1}{2}) = 1$ (maximum)
- ▶ Symmetric: $H_2(p) = H_2(1 - p)$
- ▶ Concave (diminishing returns)



An important property of entropy

Property: If X is an random variable taking n values, then $H(X) \leq \log_2 n$.

Source Coding Problem

Goal: Represent source symbols using as few bits as possible (losslessly)

Setup:

- ▶ Source alphabet $\mathcal{X} = \{x_1, x_2, \dots, x_n\}$
- ▶ Probabilities $p_1, p_2, \dots, p_n$
- ▶ Code: mapping $C : \mathcal{X} \rightarrow \{0, 1\}^*$ (binary strings)

Want:

- 1 **Uniquely decodable** - no ambiguity
- 2 **Prefix-free** (instantaneous) - can decode without lookahead
- 3 **Minimize expected length:** $L(C) = \sum p_i \cdot \ell_i$

Prefix Codes

Example: $x_1 \rightarrow 0$, $x_2 \rightarrow 10$, $x_3 \rightarrow 110$, $x_4 \rightarrow 111$

Check:

- ▶ No codeword is prefix of another
- ▶ Can decode as bits arrive

Decoding: $0110111 \rightarrow 0|110|111 \rightarrow x_1x_3x_4$

Example: $x_1 \rightarrow 0$, $x_2 \rightarrow 1$, $x_3 \rightarrow 00$, $x_4 \rightarrow 01$ not a valid coding.

Kraft – McMillan Inequality

Theorem: Let X take on the possible values $x_1, \dots, x_N$. Then, in order to be able to encode the values of X in a binary sequences (none of which is an extension of another) of respective lengths $n_1, \dots, n_N$, it is necessary and sufficient that

$$\sum_{i=1}^N 2^{-n_i} \leq 1$$

Example: $\ell = [1, 2, 3, 3]$

$$2^{-1} + 2^{-2} + 2^{-3} + 2^{-3} = 0.5 + 0.25 + 0.125 + 0.125 = 1$$

→ complete code (no wasted space)

Shannon's Source Coding Theorem

Theorem (Shannon, 1948): Let X take on the values $x_1, \dots, x_N$ with probabilities $p(x_1), \dots, p(x_n)$. Then, for any coding of X that assigns n_i bits to x_i ,

$$\sum_{i=1}^N n_i p(x_i) \geq H(X)$$

and there exists a code with:

$$\sum_{i=1}^N n_i p(x_i) < H(X) + 1$$

Interpretation:

- ▶ Entropy = fundamental limit of lossless compression
- ▶ Cannot compress below entropy on average
- ▶ Can get arbitrarily close (with block coding)

Example

For the example: $p(x_1) = 0.5$, $p(x_2) = 0.25$, $p(x_3) = 0.125$,
 $p(x_4) = 0.125$, $H \approx 1.75$ bits, $L = 1.75$ bits (optimal!)

Optimal code: $L = H$.

The Noisy Channel

Binary Symmetric Channel (BSC):

- ▶ Input: 0 or 1
- ▶ Output: 0 or 1
- ▶ Error probability p (crossover)

Simple Error-Detecting Code: Parity

Idea: Add one extra bit to make total number of 1's even

Example (4 data bits + 1 parity):

- ▶ 1011 $\rightarrow$ parity = 1 (odd number of 1's $\rightarrow$ add 1) $\rightarrow$ 10111
- ▶ 1001 $\rightarrow$ parity = 0 $\rightarrow$ 10010

Properties:

- ▶ Detects single bit errors
- ▶ Cannot correct errors (only detect)
- ▶ Rate $R = 4/5 = 0.8$

Detection: If received codeword has odd parity $\rightarrow$ error detected

Repetition Code – Simple Error Correction

Idea: Repeat each bit n times

Example (3-repetition):

▶ $0 \rightarrow 000$

▶ $1 \rightarrow 111$

Decoding (majority vote):

▶ $010 \rightarrow 0$ (two 0's, one 1)

▶ $101 \rightarrow 1$

Performance:

- ▶ Corrects up to 1 error (for $n = 3$)
- ▶ Rate $R = 1/3$ (very inefficient!)
- ▶ Better codes exist (e.g., Hamming codes)

Rate of reliable communication

Channel capacity:

$$C = 1 - H_2(p)$$

Maximum rate of reliable communication

Key insight: Can we send information reliably even with errors?

- ▶ Add redundancy $\rightarrow$ error-correcting codes
- ▶ Shannon proved: as long as rate $R < C$, reliable communication is possible

Shannon's Channel Coding Theorem

Theorem (1948): For a channel with capacity C :

- ▶ For any rate $R < C$, there exists a code with error probability $\rightarrow 0$ as block length $\rightarrow \infty$
- ▶ For $R > C$, error probability $\rightarrow 1$

Revolutionary implications:

- 1 Noise does not fundamentally limit communication
- 2 Only the **rate** matters, not the specific noise
- 3 Existence proof (non-constructive at first)

Modern codes: LDPC, Turbo, Polar codes $\rightarrow$ approach capacity

Two Random Variables

Joint entropy:

$$H(X, Y) = - \sum_{x,y} p(x, y) \log_2 p(x, y)$$

Conditional entropy:

$$H(Y|X = x_i) = - \sum_y p(y|x_i) \log_2 p(y|x_i)$$

$$H(Y|X) = - \sum_x p(x) \sum_y p(y|x) \log_2 p(y|x) = H(X, Y) - H(X)$$

$$H(Y|X) \leq H(Y)$$

Chain rule:

$$H(X, Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$$

Interpretation: $H(Y|X)$ = remaining uncertainty in Y after knowing X